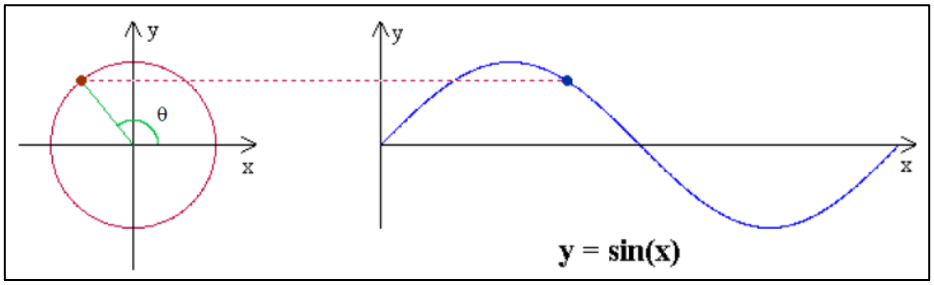
AP Precalculus
Unit 2A Day 7 Notes

Graphing Sine Functions



General form of the sine function: ____

We always plot 5 points!

- Amplitude

- Period

- Phase shift

- Vertical shift

Example 1: Graph one period of each of the following sine functions. Make sure to clearly label your axes. Show five points plotted and list all appropriate characteristics.

a. $f(x) = -3sinx$
  - Amplitude
  - Period
  - Phase shift
  - Vertical shift
  - Domain
  - Range

b. $f(x) = 10\sin\left(\frac{x}{3}\right) - 1$
  - Amplitude
  - Period
  - Phase shift
  - Vertical shift
  - Domain
  - Range

The period of a trig function is the time it takes to complete ______ cycle. The frequency is just the opposite; it's the number of complete cycles every 2π units. Because y= sinx and y = cosx cover exactly one complete cycle over this interval, their frequency is 1.

Period and frequency are inversely related. That is, the higher the frequency (more waves over units), the lower the period (shorter distance on the axis for each complete cycle).

$$\text{Period} = \frac{2\pi}{Frequency}$$

In our parent function, what represents the frequency?????

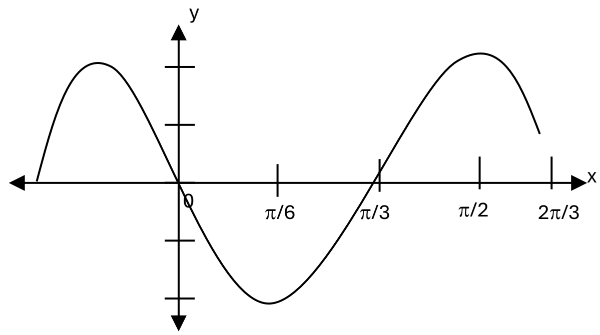
$$f(x) = a\,sin(\,bx - c) + d \qquad \text{or} \qquad f(x) = a\,sin\,\left(b\left(x - \frac{c}{b}\right)\right) + d$$

Example 2:

a. $f(x) = -2\sin\left(x + \frac{\pi}{2}\right)$
- Amplitude
- Period
- Phase shift
- Vertical shift
- Domain
- Range
- Frequency

b. $f(x) = 2\sin(3x - \pi) + 1$
- Amplitude
- Period
- Phase shift
- Vertical shift
- Domain
- Range
- Frequency

Example 3: Fill out the characteristics of the function. Then, write an equation based on the given starting points.



a. Starting at $x = \frac{\pi}{3}$

Amplitude __________ Period __________

Phase shift __________ Vertical shift __________

b. Starting at $x = \frac{\pi}{3}$

Amplitude __________ Period __________

Phase shift __________ Vertical shift __________

c. Fill out the characteristics. Then, write a sine equation to represent the function graphed.

Amplitude __________ Period __________

Phase shift __________ Vertical shift __________